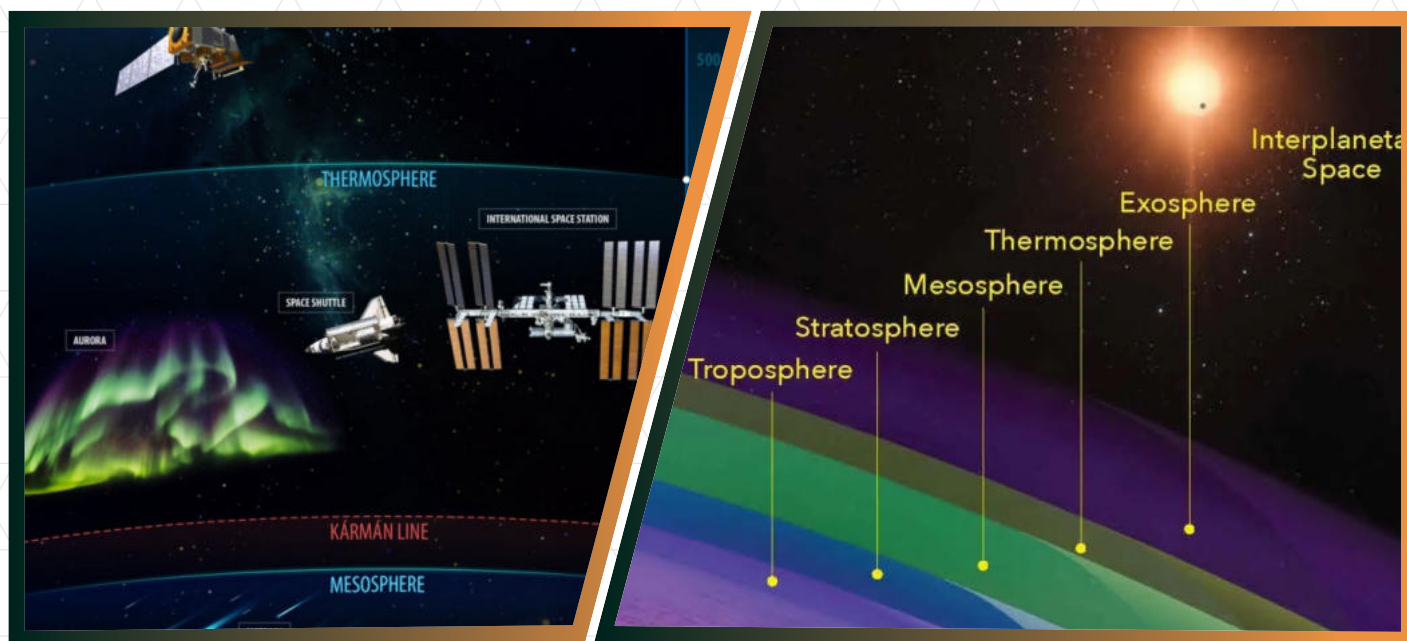




UNIT 05

ENVIRONMENTAL CHEMISTRY-I (THE ATMOSPHERE)



In this chapter, we will learn about "The Atmosphere," which is a blanket of gases surrounding the Earth. It plays a crucial role in supporting life and regulating the Earth's climate.

ATMOSPHERE

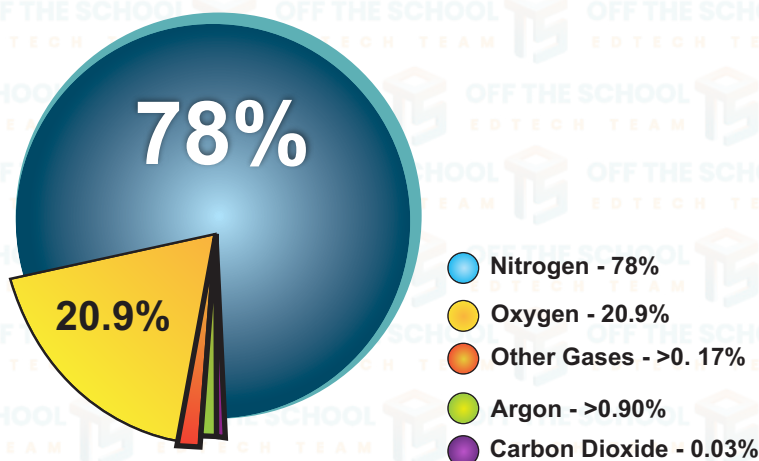
The layer of air surrounding the earth at a height of around 1600 km and it consists of many gases is known as atmosphere.

COMPOSITION OF ATMOSPHERE

The atmosphere is composed mainly of nitrogen, oxygen, and other trace gases. It also contains water vapor and suspended particles.

PERCENTAGES OF GASES IN ATMOSPHERE

Gas	Percentage in Air
Nitrogen (N_2)	~78%
Oxygen (O_2)	~21%
Argon (Ar)	~0.93%
Carbon Dioxide (CO_2)	~0.04%
Neon (Ne)	Trace Amount
Helium (He)	Trace Amount
Methane (CH_4)	Trace Amount
Others	Trace Amount



In the atmosphere, there are five distinct layers, each with unique characteristics and importance.

1. TROPOSPHERE

The layer closest to the Earth's surface where weather occurs. Troposphere constitutes the major portion of the atmosphere.

2. STRATOSPHERE

Contains the ozone layer that absorbs harmful ultraviolet radiation. Also called ozonosphere as ozone layer is found here.

3. MESOSPHERE

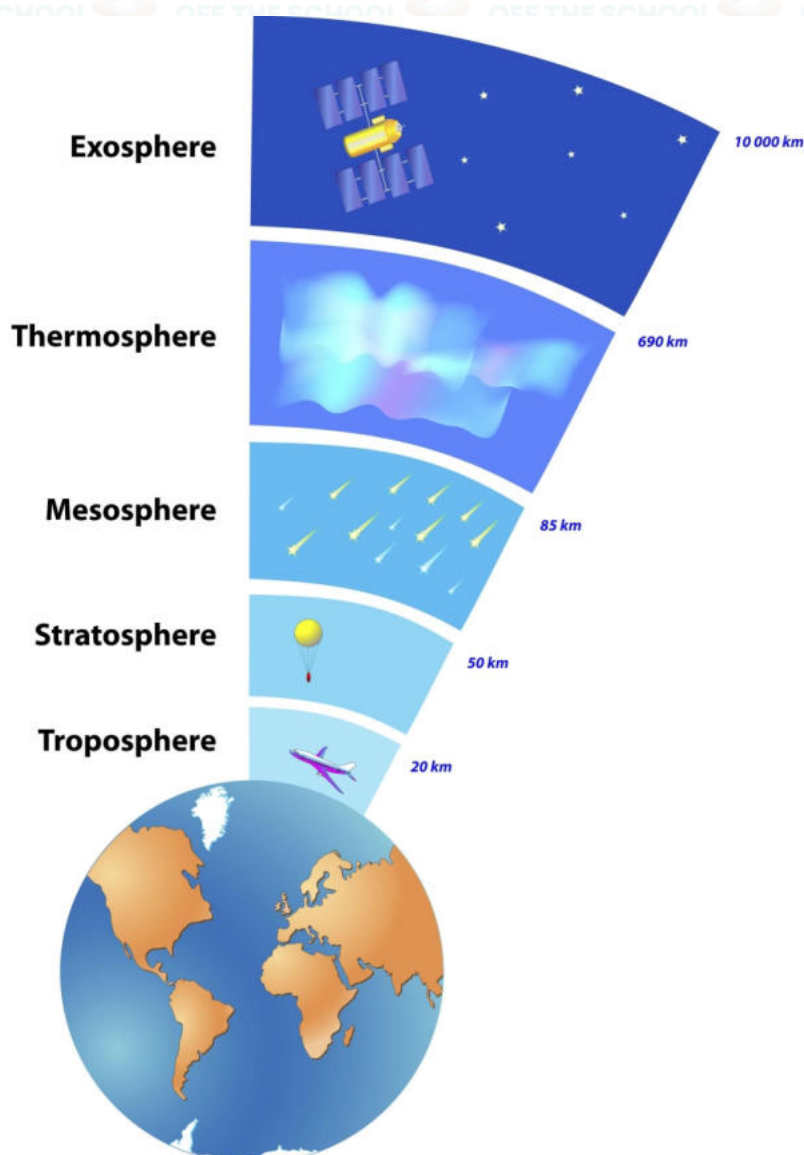
The coldest layer and where meteoroids burn up.

4. THERMOSPHERE

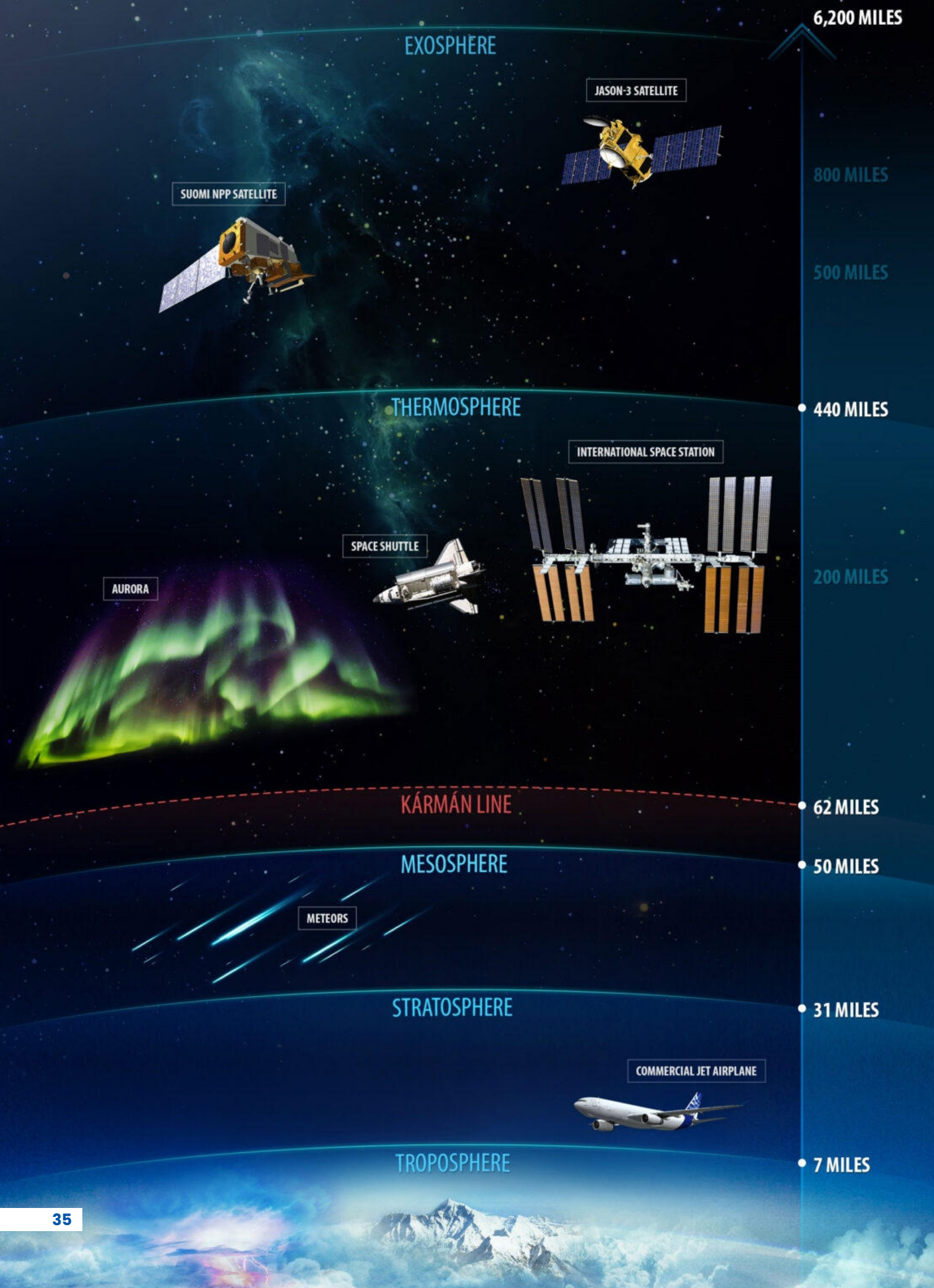
The layer where the International Space Station orbits and where temperature increases with altitude.

5. EXOSPHERE

The outermost layer, merging with space, and contains a thin dispersion of gases.



Properties	Troposphere (Tropo- means "turning" or "changing")	Stratosphere (Stratos- means "layer" or "spread out"	Mesosphere (Meso- means "middle")	Thermosphere (Thermo- means "heat") Ionosphere Exosphere
Altitude (km)	0 - 12	12 - 50	50 - 80	Above 80
Temperature range (°C)	15 to -56	-56 to -5	-5 to -93	-93 to 1800



DISTINGUISH BETWEEN TROPOSPHERE AND STRATOSPHERE

TROPOSPHERE	STRATOSPHERE OR OZONOSPHERE
1. Closest to the Earth's surface.	Located above the Troposphere.
2. Where weather phenomena occur.	Generally free of weather phenomena.
3. Temperature decreases with altitude.	Temperature increases with altitude.
4. Contains most of the Earth's atmospheric mass.	Contains the ozone layer, which absorbs UV radiation.
5. Higher water vapor content.	Lower water vapor content.
6. Lower concentration of ozone. (0.1 ppm)	Higher concentration of ozone. (10ppm)
7. Air turbulence and mixing of gases are common.	Relatively stable and less turbulent.
8. Most of the Earth's clouds and weather systems occur	Aircraft usually fly in the lower part of this layer

POLLUTANTS

Pollutants are substances or agents that contaminate the environment and can cause harm to living organisms and ecosystems.

TYPES OF POLLUTANTS

1. AIR POLLUTANTS

This is caused by the addition of undesirable substances into the atmosphere. It is of the following two types:

a) Tropospheric pollution: It is caused by gaseous air pollutants like SO_2 , NO_2 , CO_2 , or H_2S .

b) Stratospheric pollution: The damage to the ozone layer by the action of compounds like nitric acid and chloroform carbons constitute this particular type of pollution.

2. Water pollutants	3. Soil pollutants	4. Thermal pollutants
5. Radioactive pollutants	6. Noise pollutants	7. Light pollutants

MAJOR AIR POLLUTANTS

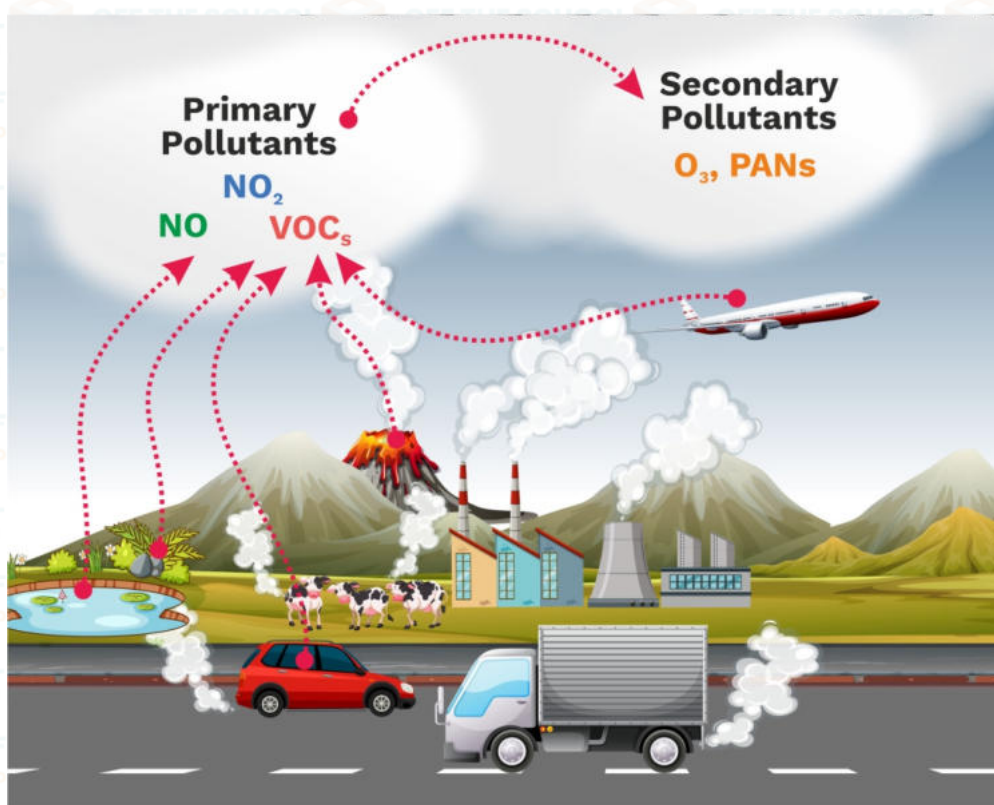
There are two types of pollutants:

1. PRIMARY POLLUTANTS

That are directly emitted into the atmosphere as a result of human activities or natural processes (SO_2 , SO_3 , CO , NH_3 , Hydrocarbon etc.)

2. SECONDARY POLLUTANTS

Which are formed by the primary pollutants (H_2SO_4 , H_2CO_3 , HCl , Ozone etc.)



Vehicles' smoke



Industrial smoke



Rotting vegetation



Smoke emitted from thermal power station



CFCs in aerosol spray



Volcanic eruption

Sources of air pollutants

Natural sources of air pollutants

MAJOR AIR POLLUTANTS AND THEIR IMPACTS

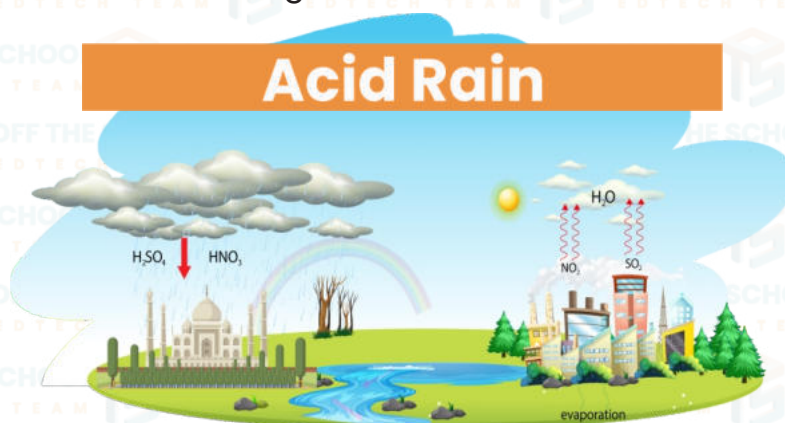
Pollutant	Sources	Environmental Risks	Human Health Risks
Carbon Monoxide (CO)	Incomplete combustion of fossil fuels, vehicle emissions	Contributes to global warming and reduces air quality	Headache, brain damage and affects the cardiovascular system.
Nitrogen Oxides (Nox)	Combustion of fossil fuels, industrial processes	Contributes to smog and acid rain, harms ecosystems	Respiratory irritant, leads to respiratory problems.

Sulfur Dioxide (SO₂)	Burning of fossil fuels, industrial activities	Causes acid rain, damages vegetation and water bodies	Irritates respiratory system, lung diseases (Bronchitis, emphysema & lung cancer).
Ozone (O₃)	Formed by reactions of NO _x and volatile organic compounds (VOCs) in sunlight	Damages crops and forests, reduces photosynthesis	Respiratory irritant, causes breathing difficulties.
Particulate Matter (PM)	Combustion processes, industrial emissions	Reduces visibility, contributes to haze and acid rain.	Respiratory and cardiovascular problems.
Lead (Pb)	Exhaust fumes from motor vehicles.	Contaminates soil and water, harms wildlife	Affects the nervous system, developmental issues.

GREENHOUSE EFFECT	GLOBAL WARMING
1. Earth has a special "blanket" of gases around it called the greenhouse effect.	Global warming means the Earth is getting warmer.
2. This blanket keeps the planet warm by trapping some of the sun's heat.	It's happening because people are doing things that add too many greenhouse gases to the air.
3. It's like when you wear a cozy jacket in cold weather to stay warm.	These gases act like too many blankets and make our planet hotter.
4. Greenhouse gases are like the threads in the blanket. They include carbon dioxide and methane.	As it gets hotter, ice at the North and South Poles melts and makes the sea rise.
5. Without this blanket, Earth would be freezing, so the greenhouse effect is good, but too much of it can make our planet too warm.	This can cause problems like floods, big storms, and changes in weather, which is not good for plants, animals, and people.

ACID RAIN

"When the pH of rain water drops below to 5.6 is called acid rain." Or It can be defined as: "Rain water having pH less than 05." Acid rain is a type of environmental pollution that occurs when acidic substances, such as sulfur dioxide and nitrogen oxides, react with atmospheric moisture and fall to the ground in the form of rain, snow, or dry particles.



EFFECTS OF ACID RAIN**1. ENVIRONMENTAL DAMAGE**

Acid rain harms aquatic life and damages soil, forests, and plants, agricultural crops

2. WATER POLLUTION

Acid rain increases the acidity of water bodies, due to acidification of soil and rocks it leaches many metals like Al, Hg, Pb and Ca then discharge them into water bodies.

3. CORROSION

Acid rain corrodes buildings, statues, and monuments made of limestone, marble, and other materials.

4. HUMAN HEALTH

Inhalation of acid rain particles can irritate the respiratory system and attacks to skin.



Fish killed by acid rain water



Effect of acid rain on trees



Stonework erosion caused by acid rain



Corrosion of metals caused by acid rain

Effects of acid rain**LIST OUT ACIDS PRESENT IN ACID RAIN**

1. Sulfuric acid (H_2SO_4)
2. Nitric acid (HNO_3)
3. Carbonic acid (H_2CO_3)

Previous Year's Questions

What is meant by acid rain and how it demolishes monuments?

[2025]

OZONE FORMATION

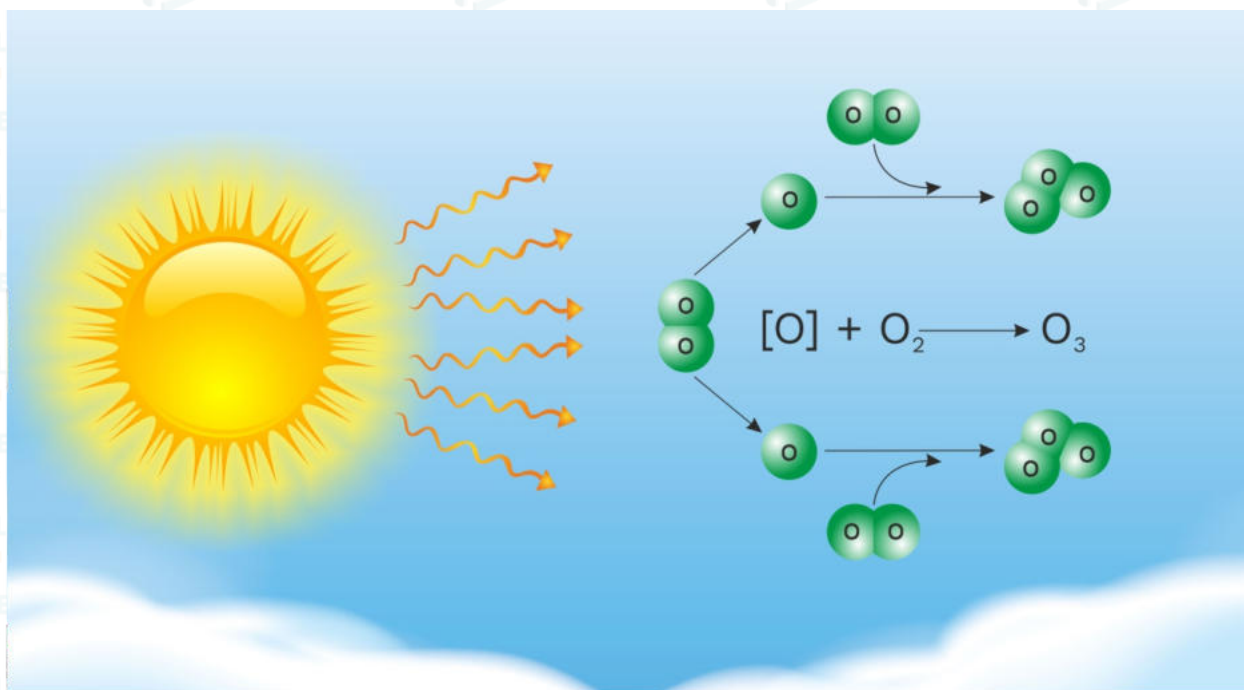
Ozone formation occurs naturally in the Earth's atmosphere through a process involving ultraviolet (UV) radiation from the sun. UV radiation splits oxygen molecules (O_2) into individual oxygen atoms (O). These oxygen atoms then combine with other oxygen molecules to form ozone (O_3).

Previous Year's Questions

Describe how does Ozone depletion occur in stratosphere by chlorofluoro carbons (CFCs) also indicate any three harmful impacts of Ozone depletion.

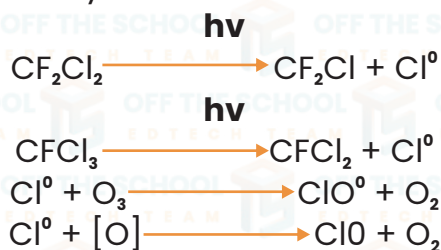
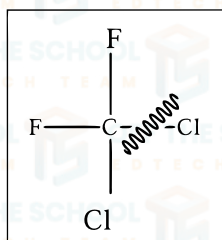
[2024]

FORMATION OF OZONE TAKES PLACE AS SHOWN BELOW



OZONE DEPLETION

Ozone depletion refers to the destruction of the ozone layer in the stratosphere due to humanmade chemicals, such as chlorofluorocarbons (CFCs). Such as Chlorofluoro carbons (CF_2Cl_2)^[1] called freons, absorb the ultraviolet radiations and get photolyzed to liberate free chlorine atoms. The free chlorine atoms catalyze the decomposition of ozone resulting in the depletion of ozone layer.



EFFECTS OF OZONE DEPLETION

1. INCREASED UV RADIATION

Thinning of the ozone layer allows more harmful UV radiation to reach the Earth's surface, increasing the risk of skin cancer and cataracts in humans.

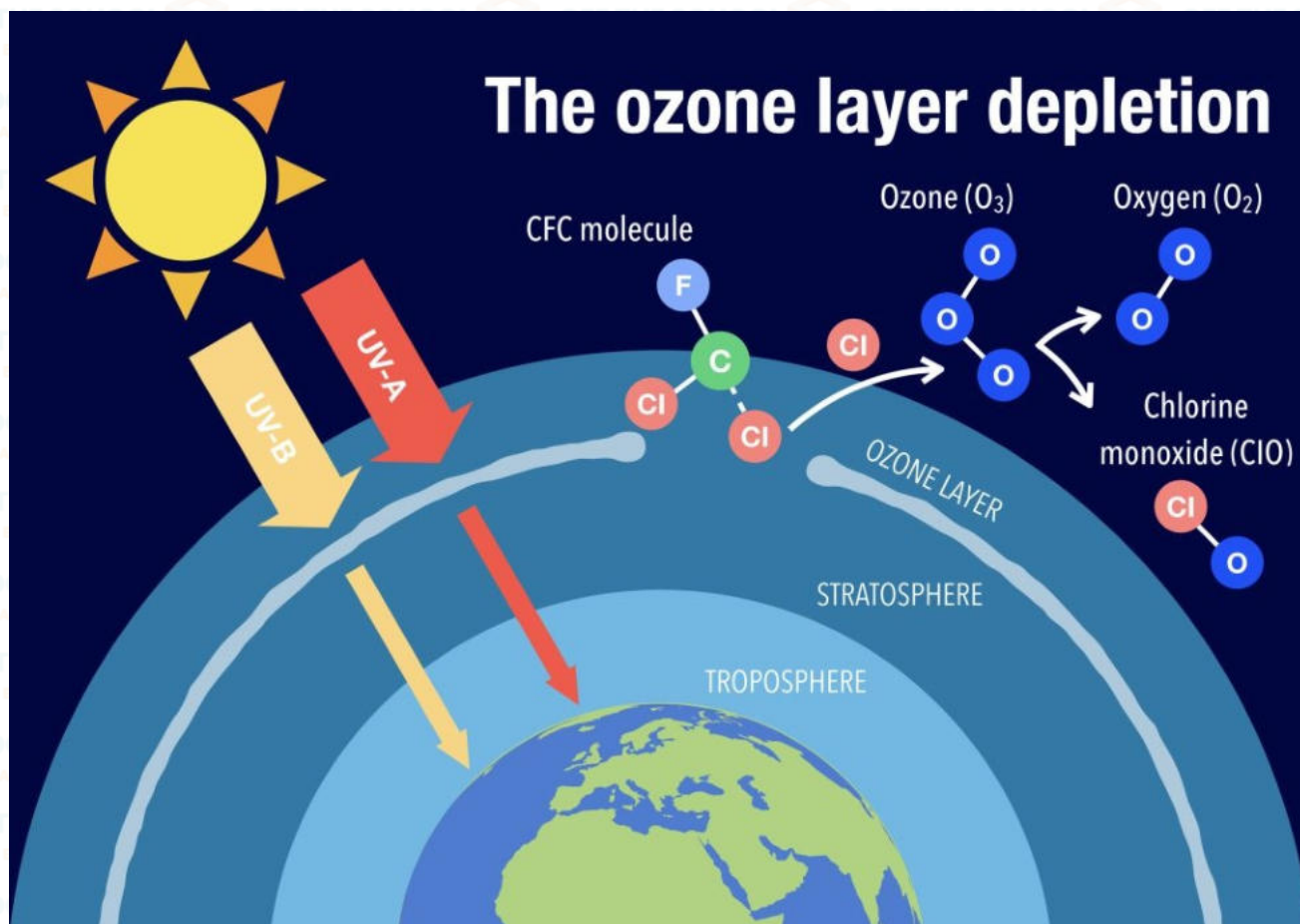
2. IMPACT ON MARINE LIFE

Higher UV levels can harm marine ecosystems, affecting phytoplankton, fish larvae, and other aquatic organisms.

3. AGRICULTURAL DISRUPTION

Increased UV radiation can reduce crop yields and disrupt ecosystems, impacting food chains.

^① One molecule of CF_2Cl_2 can destroy more than one thousand O_3 molecules in the stratosphere.



GREENHOUSE EFFECT AND GLOBAL WARMING

The greenhouse effect is a natural process that warms the Earth's surface. When the sun's energy reaches the Earth, some is absorbed, while the rest is radiated back into space as heat.

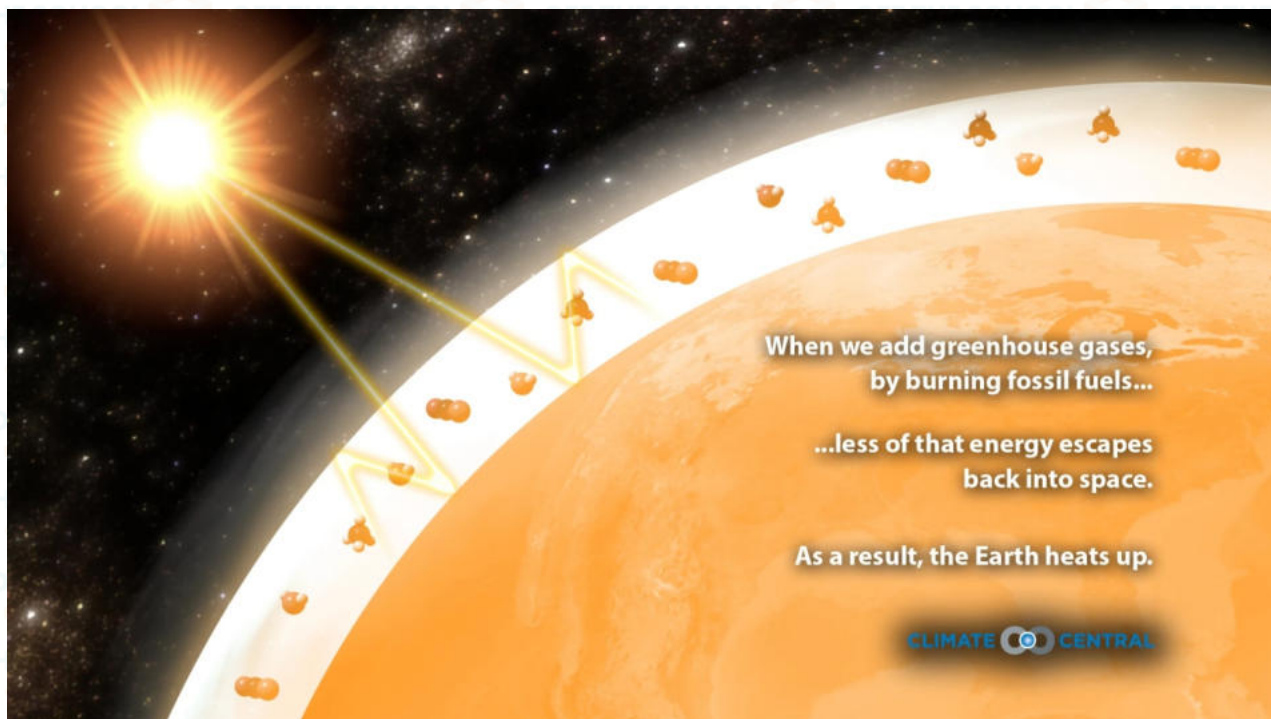
EXPLANATION: Greenhouse gases, like carbon dioxide (CO₂), methane (CH₄), and water vapor (H₂O), trap some of this heat, preventing it from escaping the atmosphere. This trapped heat warms the Earth and sustains life as we know it. However, human activities, such as burning fossil fuels and deforestation, have led to an increased concentration of greenhouse gases, enhancing the greenhouse effect and causing global warming.

The process of global warming is primarily driven by the enhanced greenhouse effect, which is caused by an increase in greenhouse gas concentrations in the atmosphere. The most significant greenhouse gas contributing to global warming is carbon dioxide (CO₂). The chemical equation describing the combustion of fossil fuels, which releases CO₂ into the atmosphere, is as follows:

COMBUSTION OF FOSSIL FUELS (E.G., COAL, OIL, AND NATURAL GAS)



This equation represents the burning of fossil fuels, which releases carbon dioxide into the atmosphere. The increasing concentration of CO₂, along with other greenhouse gases, traps heat in the Earth's atmosphere, leading to global warming and climate change.



EFFECTS OF GLOBAL WARMING

1. RISING TEMPERATURES

Global warming leads to higher average temperatures worldwide, resulting in heatwaves and extreme weather events.

2. MELTING GLACIERS AND ICE CAPS

Warming temperatures cause the melting of glaciers and ice caps, contributing to rising sea levels.

3. OCEAN ACIDIFICATION

Increased CO_2 levels in the atmosphere are absorbed by the oceans, leading to ocean acidification, which harms marine life, especially coral reefs.

4. DISRUPTION OF ECOSYSTEMS

Global warming affects habitats and can lead to changes in the distribution and behavior of species, affecting biodiversity and ecosystem balance.

LINK TO LEARNING

<https://phet.colorado.edu/en/simulations/greenhouse-effect>

LINK 01



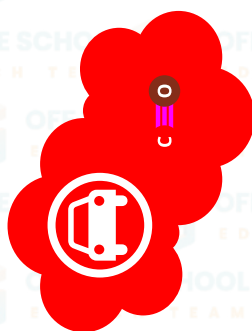
Interesting information

Greenhouse is a big room made of glass panels or transparent plastic sheets. It provides warm environment to the plants and vegetation grown inside it so that they can grow well during winter. Glass or transparent plastic sheets used in greenhouses allow the Sun's heat to enter the greenhouse and trap the heat which is reflected back by the Earth. The heat trapped by the walls and roof of the greenhouse keeps the inside environment warm.

**CO₂****Tornado****Forest Fires****Landslide****Drought****Eruption****Tsunami****Deglaciation****Effects of Global Warming**

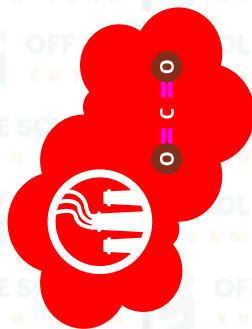
A BRIEF GUIDE TO ATMOSPHERIC POLLUTANTS

A number of different chemical entities, from a range of sources, can contribute towards atmospheric pollution, the consequences of which can include global warming and smog. This graphic looks at a selection of major groups of atmospheric pollutants, their major sources, and their effects.



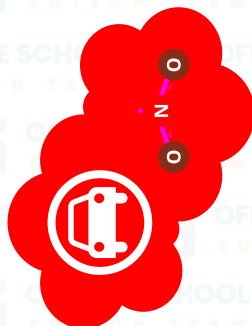
CARBON MONOXIDE

A gas generated by the incomplete combustion of fuels – primarily from road transport. Affects human health, as it reduces oxygen-carrying capacity of the blood. It also reacts with other atmospheric gases to produce ozone.



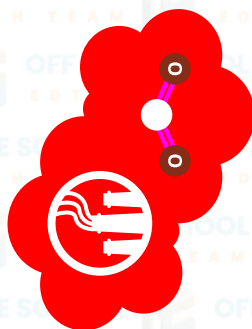
CARBON DIOXIDE

A gas generated by the burning of fossil fuels in the production of electricity. Also emitted by natural processes. Human emissions are linked with rising atmospheric CO₂ levels and anthropogenic global warming.



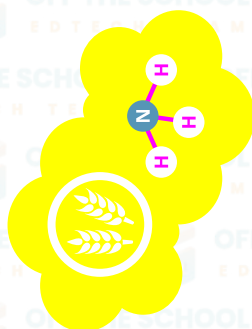
NITROGEN OXIDES

Primarily created by combustion in road transport. Nitrous oxide is an important global warming contributor, whilst nitrogen dioxide is involved in ground-level ozone forming reactions, and is also a component of smog.



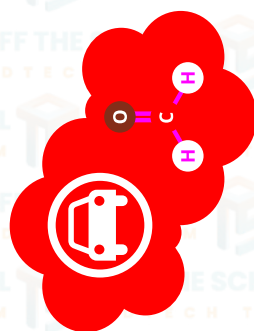
SULFUR DIOXIDE

The primary source of sulfur dioxide is the burning of fossil fuels to generate electricity. It can contribute to smog, reacts with water to produce acid rain, and can also cause wheezing and breathing problems for asthmatics.



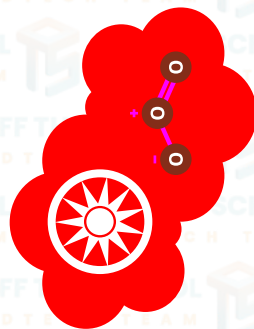
AMMONIA

Ammonia's primary atmospheric source is from its use in agriculture, such as manure & fertilisers. It can react with other pollutants to produce particulate matter. It also has the ability to over-enrich ecosystems with nitrogen.



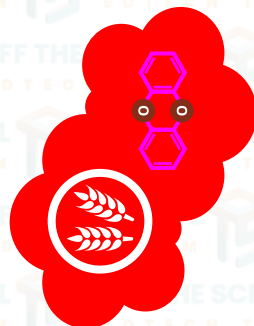
VOCs

VOCs (volatile organic compounds) are emitted naturally by vegetation. Amongst significant human sources is road transport, as well as solvents. They can contribute to formation of ground-level ozone and smog.



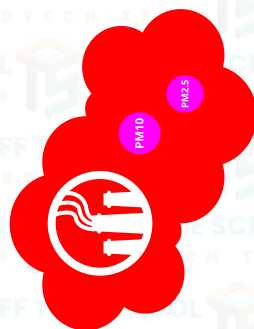
OZONE

The ozone layer shields us from UV radiation, but ground-level ozone is a major pollutant. It's formed from other pollutants in the presence of sunlight. Ozone is a major component of smog, and can also cause health effects.



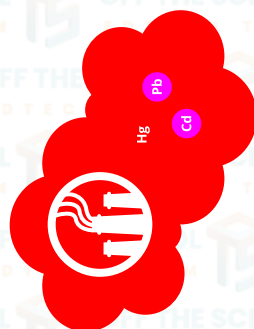
POPs

POPs (persistent organic pollutants) are volatile chemicals released into the atmosphere, often from agricultural or industrial uses. They persist in the environment and can have health effects on both wildlife & humans.



PARTICULATE MATTER

Particulate matter is composed of a huge number of different components. Some are directly emitted, while others are generated by reactions in the atmosphere. They cause haze and can also cause lung problems if inhaled.



HEAVY METALS

Heavy metals are released into the atmosphere from a range of sources, including burning of fossil fuels and road transport emissions. Some, such as mercury and lead, have toxic health effects in humans.